🌱 RDE Wild Plants - Advanced Plant Growing System

A highly optimized and feature-rich plant growing system for FiveM servers using ox_core v3, ox_lib, ox_inventory, and ox_target.

✨ Features

- 🌿 **4-Stage Growth System** - Seedling → Young Plant → Flowering → Ready to Harvest
- 🌦️ **Dynamic Weather Effects** - Weather conditions affect growth speed (Rain = faster, Thunder = slower)
- 📊 **Quality System** - Each plant has a unique quality rating affecting yield
- 🎯 **ox_target Integration** - Intuitive interaction system
- 🗄️ **Database Persistence** - Plants survive server restarts
- 🔄 **Real-time Synchronization** - Triple-redundant statebag system for perfect sync
- 🛡️ **Triple Admin Verification** - ACE permissions, ox_core groups, and Steam ID whitelist
- 🌍 **Multi-language Support** - English and German included
- 🎨 **3D Text Display** - Shows growth stage and time remaining above plants
- ⚡ **Performance Optimized** - Efficient rendering and update systems
- 🎲 **Random Events** - 5% chance of plant failure adds realism

📋 Requirements

- ox_core v3 - Character framework
- ox_lib - UI library
- ox_inventory - Inventory system
- ox_target - Targeting system
- oxmysql - Database connector

📦 Installation

1. **Download and Extract**

bash

```bash
cd resources
git clone [your-repo-url] rde_wildplants
```

2. **Add to server.cfg**

cfg

```
ensure ox_core
ensure ox_lib
ensure ox_inventory
ensure ox_target
ensure oxmysql
ensure rde_wildplants
```

3. **Add Items to ox_inventory** Add these items to your `ox_inventory/data/items.lua`:

lua

```lua
['weed_seed'] = {
    label = 'Weed Seed',
    weight = 10,
    stack = true,
    close = true,
    description = 'A seed for growing plants'
},

['harvest_tool'] = {
    label = 'Harvest Tool',
    weight = 500,
    stack = false,
    close = true,
    description = 'Tool for harvesting plants'
},

['harvested_weed'] = {
    label = 'Harvested Weed',
    weight = 50,
    stack = true,
    close = true,
    description = 'Freshly harvested plant material'
}
```

4. **Configure Admin Permissions** Edit `config.lua`:

lua

```lua
Config.AdminSystem = {
    acePermission = 'rde.weed.admin',
    steamIds = {
        'steam:110000xxxxxxxxx' -- Add your Steam IDs
    },
    oxGroups = {
        ['admin'] = 0,
        ['superadmin'] = 0
    }
}
```

5. **Database Setup** The table is created automatically on first start. No manual SQL import needed!

🎮 Usage

For Players

1. **Planting**
    - Have a `weed_seed` in your inventory
    - Use the item or trigger event: `TriggerEvent('rde_plants:plantSeed')`
    - Wait for the planting animation to complete
2. **Harvesting**
    - Wait for plant to reach stage 4 (Ready to Harvest)
    - Have a `harvest_tool` in your inventory
    - Use ox_target to interact with the plant
    - Select "Ready to Harvest" option
    - Wait for harvest animation to complete
3. **Growth Stages**
    - **Stage 1**: 🌱 Seedling (15 minutes)
    - **Stage 2**: 🌿 Young Plant (15 minutes)
    - **Stage 3**: 🌾 Flowering (15 minutes)
    - **Stage 4**: ✅ Ready to Harvest (0 minutes)

    Total: ~45 minutes (affected by weather)

For Admins

Commands:

lua

```lua
/deleteplants    -- Delete all plants (requires admin permission)
/countplants     -- Show count of active plants
/debugplants     -- (Client) Show detailed plant debug info
/debugplantsserver -- (Server) Show server-side plant data
```

Admin Verification Methods:

1. ACE Permission: `add_ace identifier.steam:110000xxx rde.weed.admin allow`
2. ox_core Groups: Admin or Superadmin with grade 0+
3. Steam ID Whitelist: Add to `Config.AdminSystem.steamIds`

⚙️ Configuration

Growth Settings

lua

```lua
Config.GrowthStages = {
    [1] = {
        model = `prop_weed_02`,
        time = 900, -- 15 minutes in seconds
        label = "seedling",
        scale = 0.1
    }
    -- ... more stages
}
```

Weather Effects

lua

```lua
Config.Growth.weatherEffects = {
    RAIN = 1.2,       -- 20% faster in rain
    THUNDER = 0.8,    -- 20% slower in thunderstorm
    EXTRASUNNY = 1.15, -- 15% faster in sun
    CLEAR = 1.1,      -- 10% faster
    CLOUDS = 1.0,     -- Normal speed
    OVERCAST = 0.9,   -- 10% slower
    FOGGY = 0.85      -- 15% slower
}
```

Harvest Rewards

lua

```lua
Config.HarvestRewards = {
    min = 25,
    max = 75,
    item = "harvested_weed"
}
```

Planting Restrictions

lua

```lua
Config.PlantingRestrictions = {
    MinDistanceBetweenPlants = 1.5,
    MaxGroundAngle = 30.0,
    MinHeightClearance = 4.0,
    CheckRadius = 2.0
}
```

Language Support

lua

```lua
Config.DefaultLanguage = 'en' -- Options: 'en', 'de'
```

🔧 Advanced Features

Triple Sync System

The script uses three redundant synchronization methods:

1. **GlobalState Statebag** (Primary)
2. **Direct Event Sync** (Backup)
3. **Manual GlobalState Read** (Fallback)

This ensures plants are always synchronized across all clients.

Ground Validation

- ❌ Cannot plant on roads, concrete, or invalid surfaces
- ❌ Cannot plant indoors
- ❌ Cannot plant in water
- ❌ Cannot plant on steep slopes (>30°)
- ❌ Must have clearance above plant
- ❌ Must maintain distance from other plants

Auto-Save System

- Plants are saved to database every 5 minutes
- Automatic save on resource stop
- All plants persist across server restarts

🐞 Debugging

Enable debug mode in `config.lua`:

lua

```lua
Config.Debug = true
```

Debug Output:

- Plant creation/deletion
- Stage changes
- Sync operations
- Admin verifications
- Database operations

Client Debug Command:

```
/debugplants
```

Shows:

- Total plants loaded
- Plant IDs with stage and entity info
- GlobalState plant count

Server Debug Command:

```
/debugplantsserver
```

Shows:

- All plants with stage, time, and owner
- GlobalState sync status

📊 Performance

- **Optimized Rendering**: Only plants within 5.0 units show 3D text
- **Efficient Updates**: Batched database saves every 5 minutes
- **Smart Sync**: Only updates when plants change
- **Model Caching**: Models are loaded once and reused
- **Entity Cleanup**: Automatic cleanup on player logout

🔒 Security

- **Triple Admin Verification**: Prevents unauthorized plant deletion
- **Server-Side Validation**: All checks done on server
- **Anti-Exploit**: Ground checks prevent invalid placements
- **SQL Injection Prevention**: Uses parameterized queries
- **Rate Limiting**: Prevents spam planting

🌍 Localization

Add your own language in `config.lua`:

lua

```lua
Config.Languages.fr = {
    success = '✅ Succès',
    error = '❌ Erreur',
    -- ... more translations
}
```

📝 Events

Client Events

lua

```lua
TriggerEvent('rde_plants:plantSeed')
TriggerServerEvent('rde_plants:createPlant', coords)
TriggerServerEvent('rde_plants:startHarvesting', plantId)
TriggerServerEvent('rde_plants:harvestComplete', plantId)
TriggerServerEvent('rde_plants:destroyPlant', plantId)
TriggerServerEvent('rde_plants:requestSync')
```

Server Events

lua

```lua
TriggerClientEvent('rde_plants:harvestPlant', source, plantId)
TriggerClientEvent('rde_plants:syncPlants', source, plants)
```

🔄 Update Guide

1. Stop your server
2. Backup your database table: `rde_wildplants`
3. Replace script files
4. Check `config.lua` for new options
5. Start server
6. Plants will automatically reload from database

❓ Troubleshooting

Plants not appearing after join:

1. Enable debug mode: `Config.Debug = true`
2. Check server console for "Loaded X plants from database"
3. Check client F8 console for "Statebag update" or "Direct sync received"
4. Run `/debugplants` in F8 to verify plants loaded
5. Run `SELECT COUNT(*) FROM rde_wildplants;` in database

Plants not growing:

1. Check server console for growth updates
2. Verify growth thread is running
3. Check if `Config.Growth.enableWeather` is true
4. Ensure plants have `timeLeft > 0`

Cannot plant:

- Check ground material (see console with debug enabled)
- Ensure not too close to other plants (1.5 units minimum)
- Verify you have a seed in inventory
- Check if ground angle is valid (<30°)

Harvest not working:

1. Verify plant is stage 4 (Ready to Harvest)
2. Ensure you have `harvest_tool` in inventory
3. Check inventory space is available
4. Look for server console errors

📜 License

This script is provided as-is. Feel free to modify for your server.

🤝 Support

For issues, please provide:

- FiveM server version
- ox_core version
- Full server console log
- Client F8 console log
- Steps to reproduce

🎉 Credits

- **Framework**: ox_core by Overextended
- **Libraries**: ox_lib, ox_inventory, ox_target
- **Script**: RDE Development

Version: 1.0.0
Last Updated: November 2025
Compatible with: ox_core v3, FiveM b3570+